

SURIN AGROMET, Thailand

WMO: 484330

Lat: 14.891N

Lon: 103.452E

Elev: 475

StdP: 14.45

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	58.3	61.0	50.7	55.7	63.7	53.1	61.0	65.6	11.8	73.1	9.9	74.9	1.6	N/A	0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	19.2	100.7	79.3	98.4	78.9	96.5	78.7	82.3	92.7	81.6	91.7	81.0	90.9	2.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.8	157.5	87.3	79.1	153.7	86.6	78.5	150.7	85.9	46.5	93.1	45.8	91.9	45.1	91.0	94.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.4	8.2	7.1	DB	53.1	102.5	2.5	2.4	51.3	104.3	49.8	105.7	48.4	107.0	46.5	108.8
			WB	50.9	85.4	2.7	3.7	48.9	88.1	47.3	90.2	45.8	92.3	43.8	95.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.0	75.5	79.6	84.2	86.7	86.0	85.2	83.9	83.4	82.7	81.7	79.6	75.7
	DBStd	4.98	4.75	5.06	4.51	3.76	2.64	2.04	2.26	1.79	2.18	2.03	3.39	4.68
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11690	792	828	1059	1101	1116	1057	1052	1037	981	982	889	796
	CDD65	6216	328	408	594	651	651	607	587	572	531	517	439	331
	CDH74	68316	2876	4355	7256	8354	7942	7206	6647	6247	5557	5044	4011	2821
	CDH80	30511	1105	2107	3860	4589	3896	3254	2763	2433	2046	1832	1563	1063
(14) Wind	WSAvg	1.6	1.8	1.5	1.6	1.4	1.3	1.5	1.5	1.5	1.1	1.6	2.1	2.5
(15) Precipitation	PrecAvg	50.7	0.1	0.4	0.9	3.3	6.8	6.5	7.3	8.2	11.2	5.0	1.0	0.1
	PrecMax	64.0	0.6	2.5	2.0	8.9	17.8	11.6	13.5	14.5	20.4	12.6	3.2	1.1
	PrecMin	38.8	0.0	0.0	0.0	0.2	1.8	3.6	3.3	2.6	5.2	1.4	0.0	0.0
	PrecStd	7.1	0.2	0.6	0.6	2.4	4.2	2.4	2.7	3.3	4.7	3.1	1.0	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.7	98.1	102.7	105.3	101.6	98.0	96.9	93.9	94.0	93.1	93.5	93.4
		MCWB	74.8	77.1	78.3	80.3	80.7	80.5	80.0	80.1	81.6	79.3	78.6	75.6
	2%	DB	90.9	95.8	99.6	101.9	98.5	95.6	94.3	92.2	91.8	91.1	91.4	90.9
		MCWB	74.2	76.0	77.6	79.9	80.3	80.4	80.2	79.8	80.5	78.5	77.5	75.4
	5%	DB	88.9	93.9	97.3	99.0	96.2	93.6	92.3	90.8	90.5	89.8	89.8	88.9
		MCWB	73.7	75.7	77.4	79.3	80.0	80.2	79.8	79.6	80.1	78.2	76.7	74.7
	10%	DB	86.3	91.1	94.9	96.5	93.9	91.8	90.3	89.4	88.9	88.4	88.1	86.3
		MCWB	72.5	74.9	77.2	78.7	79.9	80.0	79.4	79.5	79.7	78.0	76.2	73.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.3	79.7	80.4	82.7	82.7	83.0	82.7	81.9	82.8	81.3	80.3	77.8
		MCDB	88.6	93.1	95.7	97.1	94.9	93.5	92.1	90.0	90.8	89.4	90.1	89.4
	2%	WB	76.7	78.4	79.2	81.0	81.7	81.9	81.4	81.3	81.6	80.1	79.2	76.5
		MCDB	87.1	91.3	94.0	95.7	93.7	91.9	90.6	89.2	89.6	88.0	88.6	87.8
	5%	WB	75.4	77.4	78.7	80.2	81.0	81.0	80.6	80.7	80.8	79.5	78.3	75.6
		MCDB	84.8	89.8	93.1	94.4	92.4	90.8	89.3	88.4	88.4	87.0	86.9	85.9
	10%	WB	74.0	76.3	78.0	79.6	80.5	80.4	79.9	80.0	80.0	78.9	77.3	74.4
		MCDB	83.0	87.9	91.9	93.2	91.3	89.8	88.2	87.4	87.1	86.0	85.0	84.0
(35) Mean Daily Temperature Range	5% DB	MDBR	21.9	22.7	20.8	19.2	16.1	14.3	13.3	12.8	12.3	13.7	17.1	20.2
		MCDBR	22.8	24.4	23.4	22.4	19.3	16.5	16.0	14.4	14.5	15.7	18.4	21.3
		MCWBR	8.5	7.4	6.0	5.8	5.3	4.9	5.3	5.0	5.2	5.3	6.3	7.9
	5% WB	MCDBR	19.8	21.3	20.6	20.1	16.7	15.2	14.4	13.6	13.6	14.0	16.4	19.8
		MCWBR	8.1	6.9	6.3	6.1	5.5	5.3	5.6	5.3	5.4	5.3	6.1	7.9
(40) Clear-Sky Solar Irradiance	taub		0.498	0.613	0.663	0.600	0.500	0.467	0.460	0.463	0.497	0.523	0.475	0.466
	taud		2.032	1.758	1.689	1.872	2.189	2.291	2.309	2.291	2.185	2.081	2.194	2.163
	Ebn at Noon		252	228	221	236	256	262	265	267	259	248	257	258
	Edh at Noon		53	72	79	66	47	42	42	43	48	52	44	45
(44) All-Sky Solar Radiation	RadAvg		1589	1664	1704	1823	1801	1699	1611	1555	1485	1566	1570	1537
	RadStd		86	69	63	73	96	65	102	95	98	104	72	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	+0.47	N/A	N/A	N/A	N/A	+0.45	+0.40	N/A	N/A	+185

Nomenclature: See separate page